Prototyping 2

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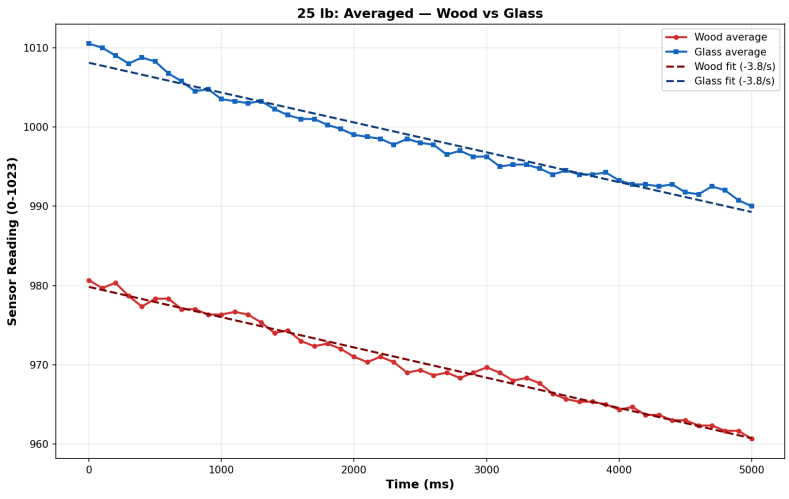
For this prototyping phase, there are three main tasks at hand:

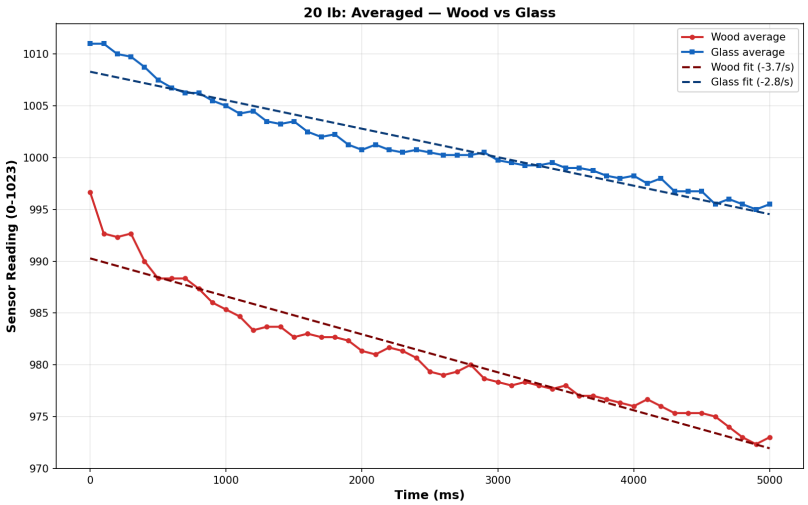
1. See if the foam's "drift" is actually drift or if there is leakage
2. See if covering up the FSR in foam increases the range
3. Research different "filters" for getting rid of noise

The first task was actually pretty simple. I already had the calibration data for the foam from last time (see **here**) so all I needed to do was redo it while putting the foam on an insulator. For this, I just simply redid some of the trials on a glass table (glass is a perfect insulator). Redoing every trial is redundant, so I just did the 20lb and 25lb trials.

The procedure was the exact same as last time. I taped down the foam sensor, put a hockey puck on top of it (to imitate a heel), placed the weight on the puck, waited around half a second (to avoid the sudden "jerk" or change in value right when you put it on), hit go, and let my computer keep track of the readings for the next 5 seconds (see previous document for the code). I did each trial 3 times. To see the data, click here.

This produced very interesting results. First off, the foam on the glass vs the wood had different starting points. On the glass, the resting point was around 1010, while on the wood the resting point was around 980.

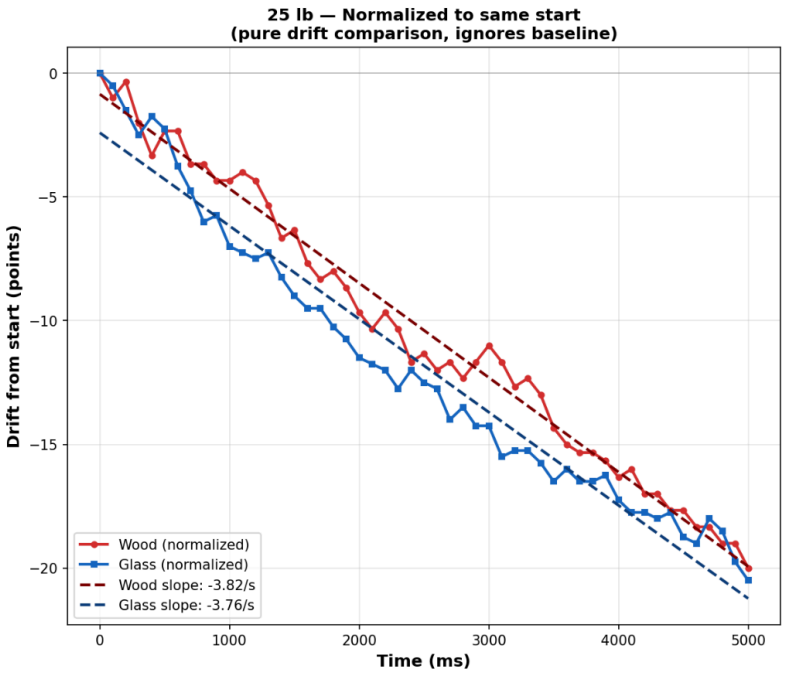


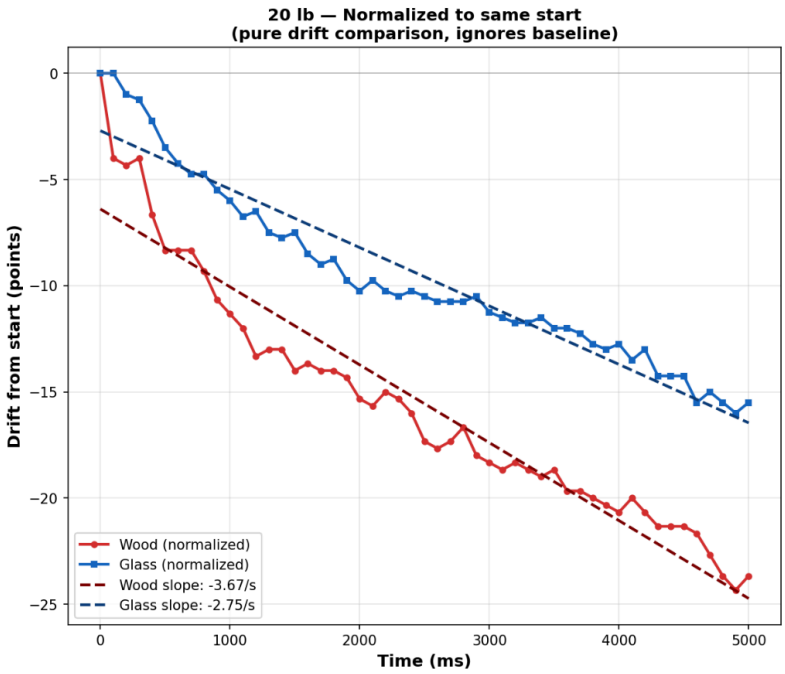


I honestly do not know why this is, but I do have some guesses:

1. The glass trails were tested a few days after I did the wood trials, so maybe the foam's resting point changes over time
2. Something in the glass changed something in the electrical current which made the resting point a little higher
3. Slightly different positioning of the tape on the foam

So, knowing that there are different starting points, I first normalized the curves (made them start at the same point) before I analyzed it

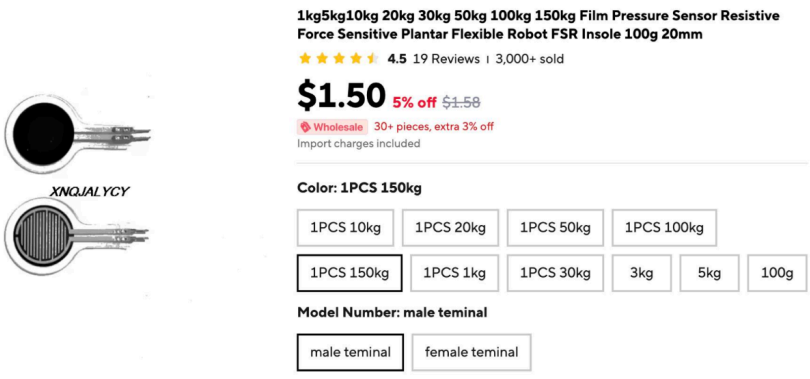


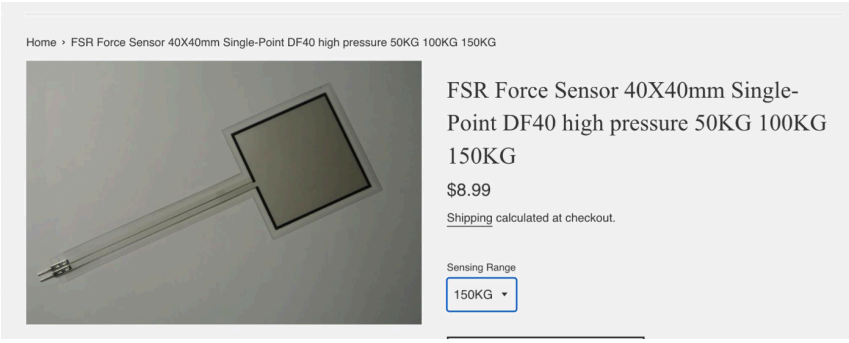


Result: Now that they are both normalized, we can compare best fit slopes. At 25 lb, wood and glass drifted identically (−3.8/s both). At 20 lb, glass drifted slightly less (−2.8 vs −3.7/s). Given these values, we can take away that there is probably a small possible insulation effect, but it is inconsistent across weights so the drift appears to be mostly real foam creep, not leakage.

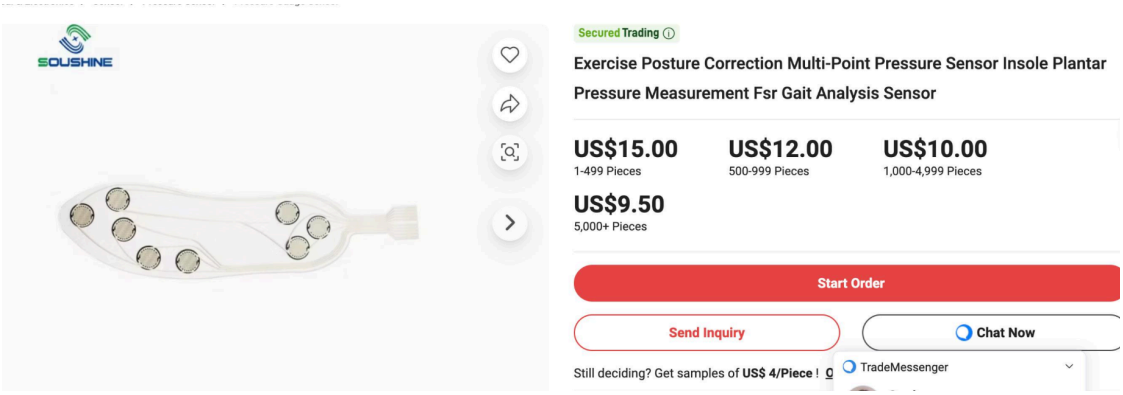
FSR Testing

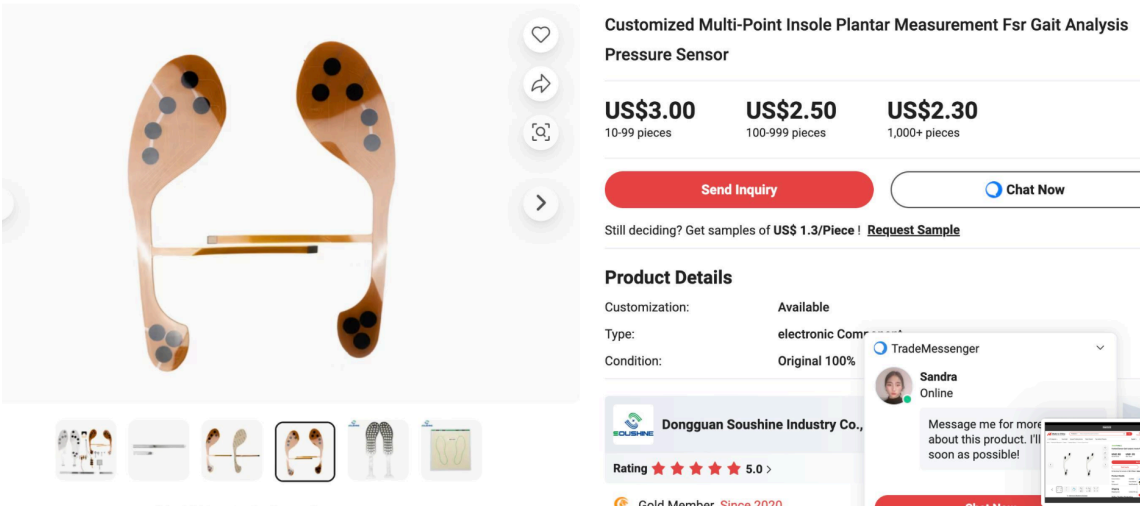
Now before we start testing, Dr. Jenkyn wanted me to do research on FSRs that can hold more weight. So I found these:

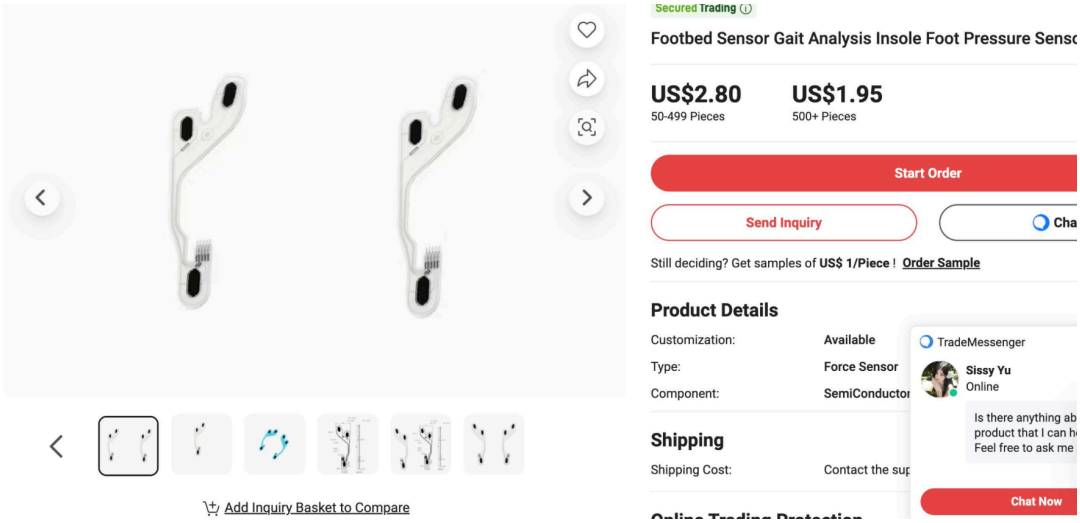




I also found a Chinese website that sells insoles with sensors already in them:







But before buying any of these, I want Dr. Jenkyn's input

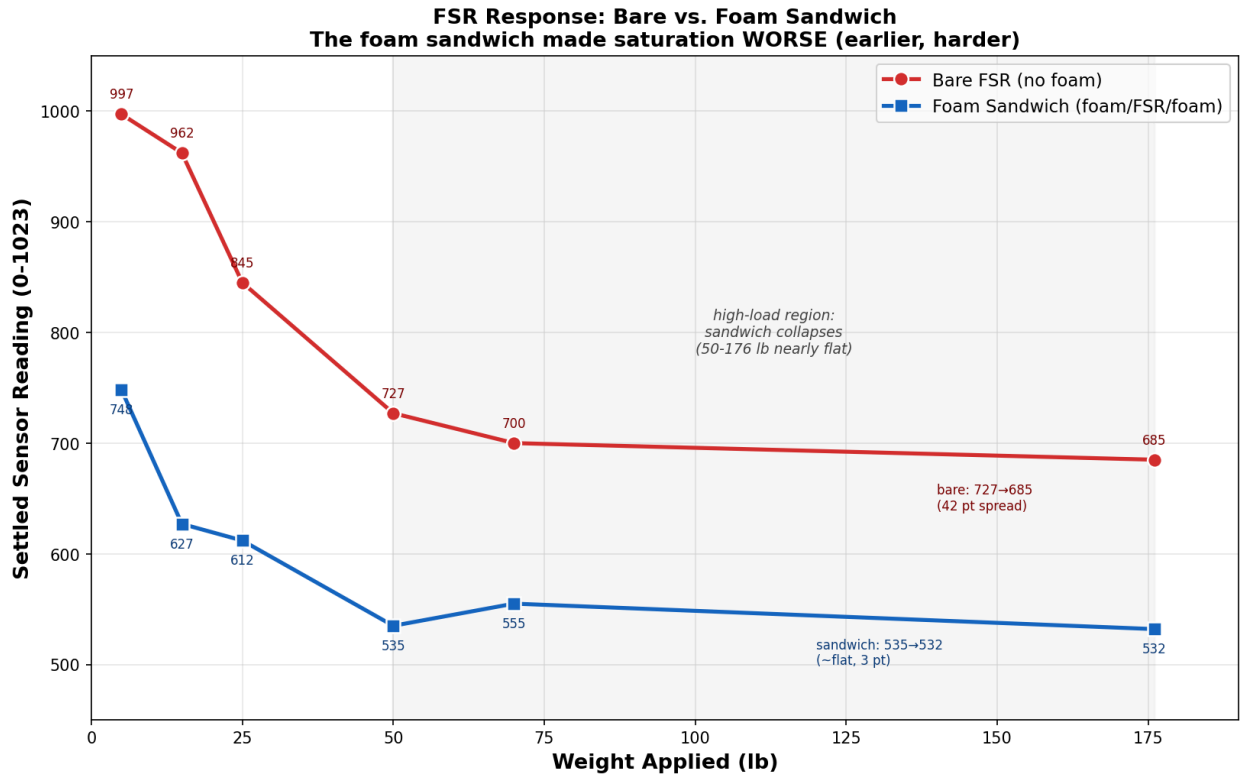
Now, back to testing:



Currently I have an FSR that has a weight bearing limit of around 22lb (what Amazon says). So, the hypothesis is this: If we cover the FSR in foam, then the load will be distributed out more and it can take more weight. To test this, I basically made two contraptions, and calibrated each using 5lb, 15lb, 25lb, 50lb, 70lb, 176lb (me standing on it). Calibration with the FSR is the same as the foam. I would place the weight on the puck, wait half a second, press go, and my computer recorded the reading for 5 seconds.



HERE IS THE DATA



This is very interesting, as it shows the exact opposite of what we were expecting. The foam sandwich made the saturation WORSE. For example, on the bare FSR, you should tell the difference between 70lb and 50lb, but you cannot on the Foam Sandwich. My best guess is that, with the puck, the foam distributed the load more evenly and completely onto the FSR surface.

Conclusion: Our hypothesis did not work. Creating a foam sandwich does not increase the range.

Noise Filters

Box / Rectangular Window

This is what I used. Every single reading is weighed equally and averaged, but lets through more “leakage”

Bartlett (Triangular) Window

The weights ramp up to a peak at the center of the window and back down, with the endpoints weighted at zero (or nonzero). This means the center reading matters most and the edges barely count. It's a gentle step up in quality from the box window.

Hann and Hamming Windows

These use smooth cosine-shaped weighting curves rather than a flat or triangular shape. They remove noise more cleanly than the box or triangle while better preserving the true shape of the signal. Haming is slightly better at canceling noise close to the signal's frequency, while Hann is better at suppressing noise farther away.

Break Frequency and Walking Speed

Walking happens at roughly 5 Hz, so the filter's cutoff needs to sit above ~5 Hz to preserve the real walking motion, while still being low enough to cut out the higher-frequency electrical noise.

Hardware Filters

It is a good idea to look into hardware filters as they filter noise before it is digested into the computer. But I think we should do this later when we have the hardware finalized and all set up.

Conclusion: Honestly, there is no single best filter for this project, I think a combination of them will work best. I want to test all of them once we get the sensor finalized to find the best combination.

Overall, I think this was a good prototyping phase. We disproved a lot of our hypotheses. I think the best way to move forward is to:

1. Get FSRs that can handle more weight and test them. We can even test them by arranging them in specific spots on an insole.
2. Continue to test the foam. Make a new foam contraption in the shape of an insole and test that.
3. Move toward lab validation



Side note: I also have a 3d printer and 3d printed an insole. This will be helpful for future testing